Type2 support

CXL.mem status and future CXL.cache

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CXL Type2 support

Agenda:

1. Summary of current Type2 support patchset under review

2. CXL cache: what we need? Discussion and potential roadmap

CXL Type2 support: an update of current ongoing work

Type2 support use case:

- Network device using CXL.mem (and CXL.cache)
- PIO buffers: driver writing the packet to device buffers instead of relying on DMA(s) for descriptor plus DMA(s) for packet.
- PIO buffers: PCI aperture ---> CXL.mem region
- CXL.cache: not enabled yet. “Complicated”

CXL Type2 support: an update of current ongoing work

Type2 support needs:

- BIOS/UEFI not touching CXL registers: EFI_RESERVED_TYPE
- Accel drivers vs CXL PCI mem class driver
- Exporting an API for accel drivers to do CXL initialization
- Fixing Type3-only code
- Adding support for creating CXL regions by kernel code/accel drivers vs user-space intervention
- Precluding CXL region being a DAX region (default for Type3)

CXL Type2 support: an update of current ongoing work

Type2 support patchset under review:

- A good part of it based on Dan Williams' previous work
- RFC:
  - Concern about BIOS and Type2: EFI_RESERVED_TYPE expected.
  - client based on a emulated Type2 device: OK but ...
  - Accel driver modifying cxl_dev_state: trusting accel drivers?
  - No CXL.cache support

Type2 support patchset under review:

- V2:
  - Exporting CXL API for accel drivers CXL initialization: opaque cxl_dev_state
  - Using SFC network driver as the client instead of emulated device
  - Moving just required header contents instead of full header files.
  - Type2 vs Type3 when CXL registers discovery: mandatory capabilities check
  - Some patches too big/complex

Type2 support patchset under review:

- V3:
  - API exported dropping the accel prefix: type3 will use it
  - Capabilities field added.
  - CXL registers discovery: no fatal if not found. No checking what is there.
  - New API function: checking registers found vs expected device capabilities (driver knows).
  - Factoring out single steps from complex patches.
  - Races when device creation: memdev vs endpoint port.

Type2 support patchset under review:

- Coming V4:
  - Build fixes: kernel configs without CXL …
  - Proper handling of sysfs files
  - Device creation races: corner case? Investigation needed
  - Avoid sfc netdev driver changes mixed with CXL core changes in the same patch.
  - …

Type2 support: following work?

- CXL reset: keep configuration or/and unroll commits
- EFI_RESERVED_TYPE: what if the BIOS does not do the right thing?
    - DAX will be created.
    - Add action by the driver for removing it (CXL configuration?)
    - Nowadays an accel driver can know if it happens, but not a way for telling the DAX subsystem “do not put your dirty hands on my memory!”

Type2 CXL.cache

- What should the kernel expect/know/do?
- Root Complex resources: snooping cache size
- Switch support? If not like the Root Complex, maybe not as critical as for Type3 (initially)
- CXL.mem + BI (device snooping control/management)
- CXL.cache IDs
- Development + testing: emulation effort?
- Virtualization support?

Type2 CXL.cache

- What should the kernel expect?
  - L2/L3 means physical addresses. Not different to current cache injection (Intel DDIO, AMD SDCI, ...)
  - Host memory cached by device:
    - Initialization/Security: ATS/IOMMU (mandatory? Same like IOMMU on/off?)
    - Root Complex snooping resources: checking availability? Snooping hints to device?
    - DMA-IOMMU-like allocation with wrapper through CXL code?
    - An PCIe TLP with ATS specifically routed and source checked ... specific CXL configuration?
  - Device memory cached by Host (using CXL.cache for coherency):
    - Anything to do? The device already knows the physical addresses ...

Type2 CXL.cache

- Root Complex resources:
  - snooping cache size. Restricting total device caches to that size. Right?
  - But we have hints: Cache_SF_Coverage in DVSEC CXL Control register, and could it be 00h (no snoop coverage), and Cache_Enable=1?

Type2 CXL.cache

- CXL.mem + BI
  - Replaces CXL.cache for allowing Snooping by the device on its HDM range.
  - Kernel:
    - Management for switching/routing/commits
    - Information/Visibility
    - What else?

Type2 CXL.cache: actions?

- Documentation/Education: how it should work from kernel PoV. Plumbing!
    - Evolving doc. Monthly meetings? Kernel impact beyond CXL specifics.

- CXL Root Complex CXL.cache discovery (switches?)
- Accel drivers API: only checking capabilities and resources. Beyond that, EXPERIMENTAL CONFIG flag?
- Host memory allocation for device caching?

- Emulation: discovery and configuration. Functionality?
    - Time to push QEMU (multilevel) cache emulation?